

# Smart Port & Maritime Congestion Predictor

SIM Graduation Project

A system that predicts vessel congestion at Egypt's main maritime gateways hours ahead, learning from live and historical ship-movement data.

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# The Problem

Egypt's maritime gateways are among the busiest chokepoints on earth. When too many vessels arrive at once, ships queue at anchorage waiting for a berth or a canal convoy slot.

⚠ **Real impact:** Suez Canal median waiting time reaches **5.89 days** during peak congestion periods.

## The Cost

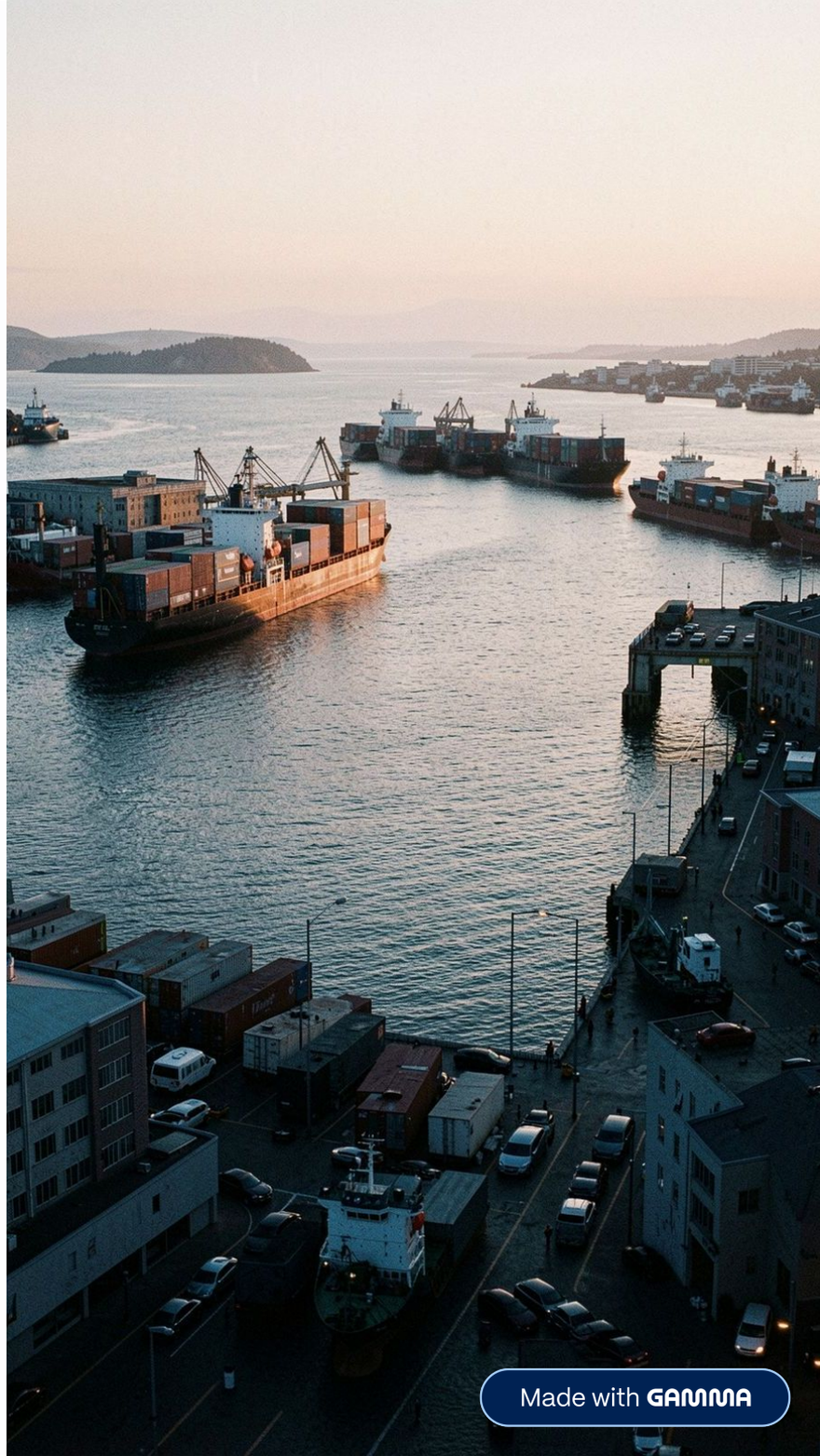
Demurrage and detention charges, wasted fuel, missed downstream logistics

## The Gap

Operators lack a reliable forward-looking signal for when congestion is about to build

## Today's Approach

Reactive management after bottlenecks form — not proactive planning



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# The Solution

A prediction engine that forecasts near-term congestion — expected vessel waiting time and a congestion index — for the Port Said / canal-approach anchorage.

## What It Learns

Patterns that precede a build-up: arrival rate climbing, anchorage filling, approach speeds dropping

## What It Predicts

The next few hours of congestion — waiting time and a congestion index — hours before it forms

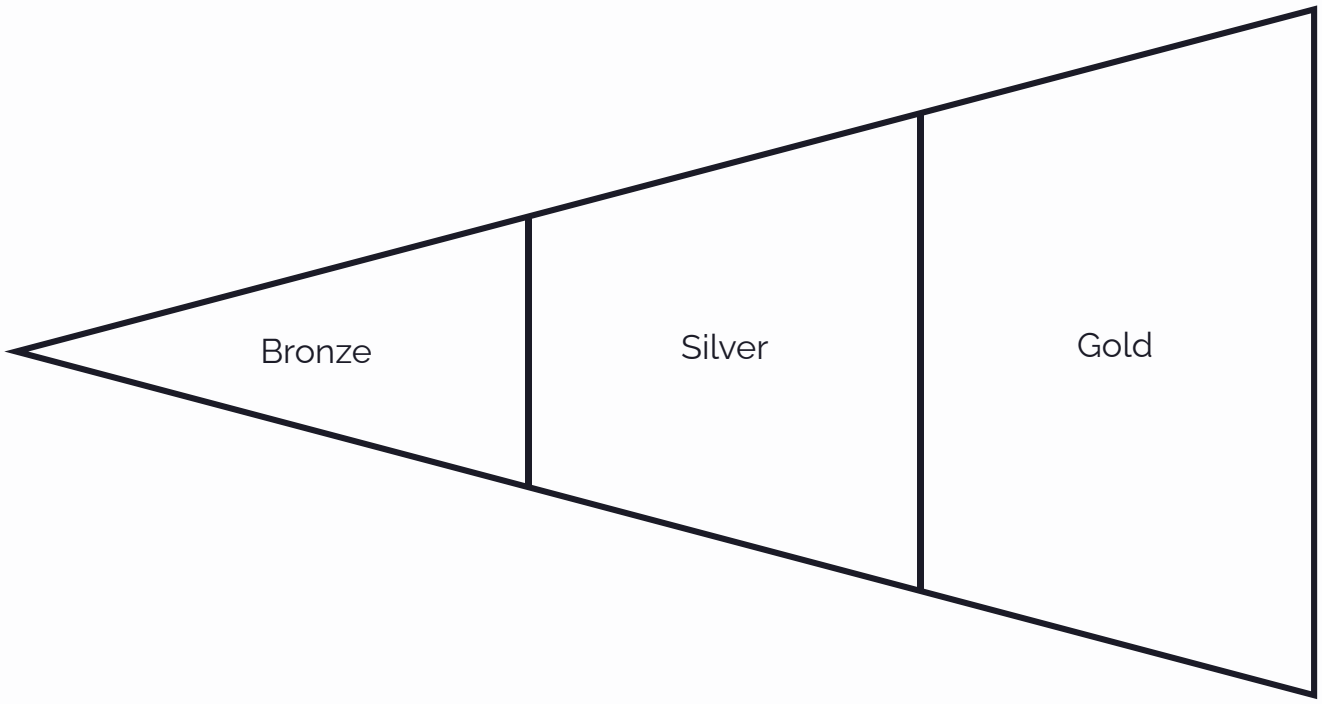
## The Key Insight

Derive both features and ground-truth labels directly from AIS vessel-tracking data — no proprietary port-authority data required

**Why it's buildable:** Using published methods (DBSCAN + convex hull to separate anchorage from berth zones, then AIS navigation status + speed + timestamps to compute waiting and turnaround times), the data is 100% real and fully obtainable by a student team.

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# How It Works: The Pipeline




### 1 Streaming Path

Live AIS → Kafka → Spark Structured Streaming computes real-time features (rolling arrival rate, vessels at anchor, mean approach speed)

### 2 Batch Path

Historical AIS → Airflow ETL → same feature logic → model training and scheduled retraining

### 3 The Bridge

A feature store serves identical feature definitions to both training and live inference, eliminating train-serve skew

### 4 Guardrails

Drift monitoring watches prediction quality and can trigger an automatic rollback to the previous model version

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# Tech Stack



## Live Ingestion & Streaming

Python WebSocket consumer (aisstream.io) → **Apache Kafka** → **Spark Structured Streaming / PySpark**



## Storage & Orchestration

**PostgreSQL** + Parquet/object storage; optionally **Cassandra** for high-write raw AIS. **Apache Airflow** for batch ETL, backfill, retraining.



## ML & Feature Store

**Feast** (anti-skew centerpiece) · **XGBoost** baseline · optional **LSTM** · scikit-learn DBSCAN · shapely / geopandas



## Tracking, Serving & Monitoring

**MLflow** · **FastAPI** (containerized) · **Evidently AI** · **Great Expectations** · Elasticsearch + Kibana / Grafana · Docker / GitHub Actions

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# Data Sources

## Streaming (Real)

**aisstream.io** — free live AIS over WebSocket, bounding-box-filtered to Port Said / the canal approach / Red Sea + Med entry. Feeds Kafka continuously.

## Batch / Historical (Real)

- Self-collected AIS — start streaming into Bronze now; accumulate genuine history over the project months
- Kaggle historical AIS as backfill — verify coverage of the Suez / Red Sea / Med region
- Open-Meteo (free) for weather features
- Optional: paid backfill (Datalastic / DataDocked, ~€80/mo) only if historical depth is needed fast

**Testing & Demos:** Replay simulator (uses REAL data) replays recorded AIS back into Kafka as if live — test the pipeline anytime, reproduce past congestion events, and run reliable live demos on defense day.

## Why This Works

All data is free, legal, and publicly available. The self-collection strategy means the team builds genuine historical depth over the project timeline — no proprietary dependencies, no licensing barriers.

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# What Makes It Technically Strong

1

## Feature Store as Anti-Skew Centerpiece

The single most defensible MLOps decision — the same feature computed once and served to both training and inference

2

## Closed Monitoring → Rollback Loop

Drift detected → alert → MLflow rolls back to the prior model version automatically

3

## Data-Quality Gates

Great Expectations in the Airflow DAGs catch bad AIS batches before they poison training

4

## Reproducible Defense Demo

Replay simulator enables testing and demos anytime — reproduce past congestion events on demand

5

## Backtesting on Real Events

Validate against known historical congestion — e.g. the 2021 Ever Given blockage aftermath

6

## Explainability

SHAP shows why congestion is forecast — exactly what real operators want to see

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# Business Model & Market

## Who It's For

### Shipping Agents / Freight Forwarders

Early warning to reschedule and cut waiting charges

### Terminal / Port Operators

Forecast to allocate berths and labor proactively

### Shipping Lines / Carriers

Slow-steam or re-sequence to avoid idle waiting

### Logistics & Import/Export Companies

Plan trucking and storage around realistic ETAs

## Revenue Models

- **B2B SaaS subscription** — tiered by port/volume
- **API / usage-based** — predictions fed into customer TMS/ERP
- **Freemium** — free basic index; paid alerts
- **Data-as-a-service** — sell cleaned AIS-derived index
- **Licensing** — custom enterprise/gov

**The value hook:** Demurrage and detention are large, direct costs. A few hours of advance warning can avoid thousands of dollars per vessel. **Competitive wedge:** Egypt/Suez-tuned models, Suez convoy dynamics, Arabic interface, and local-market pricing vs. global enterprise contracts.

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# Risks, Scope & Team

## Risks (Stated Openly)

### → Single Free-Data Dependency

Mitigated by resilient consumer + replay + Bronze persistence

### → No Official Port-Authority Data

AIS-derived-label approach is the answer; real-world validation would eventually need a partner

### → Regional Feed Coverage

Verify steady AIS traffic over target zone early in the project

## Team & Scope (6 People)

**Timeline:** Build a thin working slice end-to-end first (one port, clean pipeline), then add stretch goals (drift→rollback, SHAP, LSTM, backtesting). Realistic for six people — components are cleanly divided.

### 4 Data Engineers

Batch ETL · Streaming · Storage · Geospatial feature engineering · Orchestration

### 1 MLOps Engineer

Feature store · Model registry · Serving · Monitoring → rollback loop

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# Why This Matters

**The novel contribution:** A localized, AIS-only congestion-prediction system for Egyptian maritime gateways, delivered with production-grade MLOps — combining a real Egyptian problem with a modern, defensible data + ML pipeline.

| | |
|---|---|
| **Real Problem**<br>Egypt's maritime gateways are critical global chokepoints; congestion costs thousands per vessel per day | **Real Data**<br>Free, legal, publicly available AIS; self-collected history; no proprietary dependencies |
| **Real Business**<br>Clear customer segments, concrete value (avoided demurrage), and a defensible local moat | **Real Engineering**<br>Feature store, model registry, drift monitoring, and rollback — production-grade MLOps from day one |

**Success Metrics:** Model MAE on predicted waiting time · Prediction lead time (hours ahead) · Business impact: % reduction in demurrage in a pilot, berth-utilization improvement, ETA-reliability gain.

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